

XGBoost

Introduction:

⇒ XGBoost is not an algorithm, it's a library based on Gradient Boosting.

⇒ Why Gradient Boosting?

① Flexibility: it allows flexibility on the loss function, as it just ask to be differentiable.

② Performance: Very good performance.

③ Robust: it gives robust results.

⇒ XGBoost Features:

① Flexibility:

(i) Cross Platform

(ii) Multiple language Support

(iii) Integration with other libraries and tools.

(iv). Support all kinds of ML problems.

② Speed:

(i) Parallel processing for tree building

(ii) Optimised data structures (column block)

(iii) Cache awareness (to build decision trees)

(iv). Out of core computing

(v). Distributed computing

(vi). GPU Support

allows data to be loaded in chunks

allows training and in diff. machines

③ Performance:

(i) Regularized learning objective

(ii) Handling missing values

(iii). sparsity aware split finding

(iv). Efficient split finding

(v). Tree Pruning

XGBoost for Regression:

⇒ Suppose we have a dataset like this:

cgpa	Package
6.7	4.5
9.0	11.0
7.5	6.0
5.0	8.0

XGBoost works similar to Gradient Boost, it means it uses the same approach of stages. where the first stage be the mean and subsequent steps be the decision trees.

Since, the XGBoost uses bit different decision trees.

⇒ But how this different decision tree is formed?

stage 1: Mean of output column.

$$\text{Mean} = \frac{4.5 + 11.0 + 6.0 + 8.0}{4} = 7.3$$

Then we calculate the residuals as actual - mean. So, the dataset becomes:

cgpa	Package	model	residual
6.7	4.5	7.3	-2.8
9.0	11.0	7.3	3.7
7.5	6.0	7.3	-1.3
5.0	8.0	7.3	0.7

Now, we will keep these residuals in one mode and will find the similarity score of them.

$$\% \text{ Similarity Score (SS)} = \frac{(\text{sum of residuals})^2}{\text{No. of residuals} + \lambda}$$

where, λ = reg. parameter

For our data, $\lambda = 0$

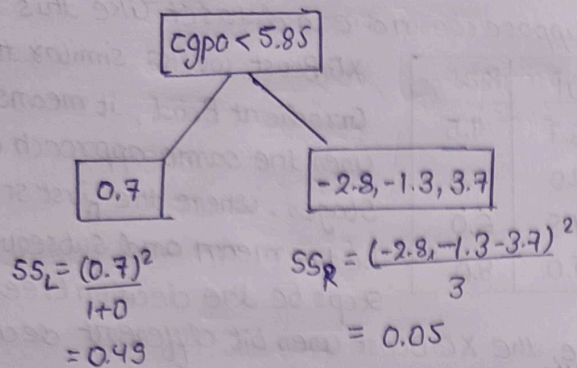
$$\% \text{ SS} = \frac{(0.3)^2}{4} = 0.02$$

Now, we will split residuals using the cgpa col. For which we will first sort the data, and find mean of each two rows.

5.0	→ 5.95
6.7	→ 7.1
7.5	→ 8.25
9.0	→ 8.25

3 different splitting criteria

Now, splitting criteria ① $\rightarrow 5.85$

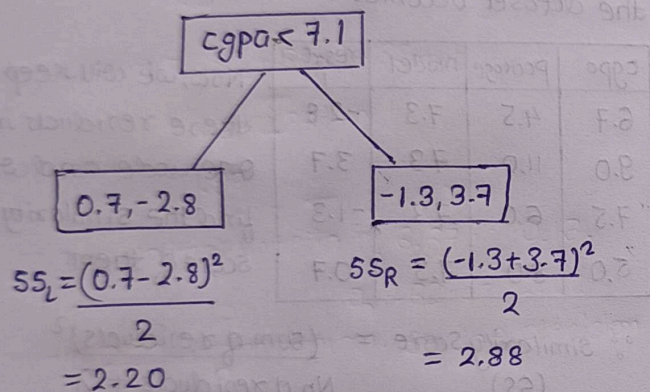


Now, we will calculate the gain for this split,

$$\text{gain} = (SS_L + SS_R) - SS_{\text{root}}$$

$$= (0.49 + 0.05) - 0.02 = 0.52$$

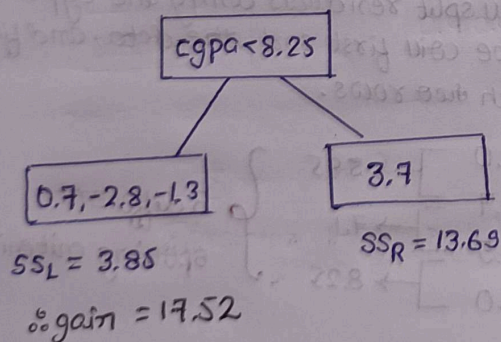
Now, splitting criteria ② $\rightarrow 7.1$



$$\therefore \text{gain} = (SS_L + SS_R) - SS_{\text{root}}$$

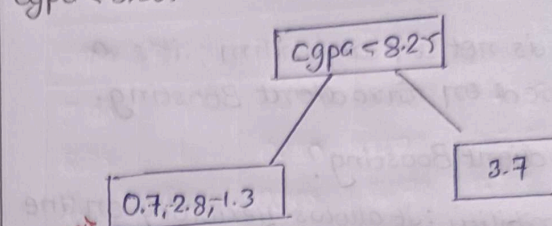
$$= 5.06$$

Now, splitting criteria ③ $\rightarrow 8.25$



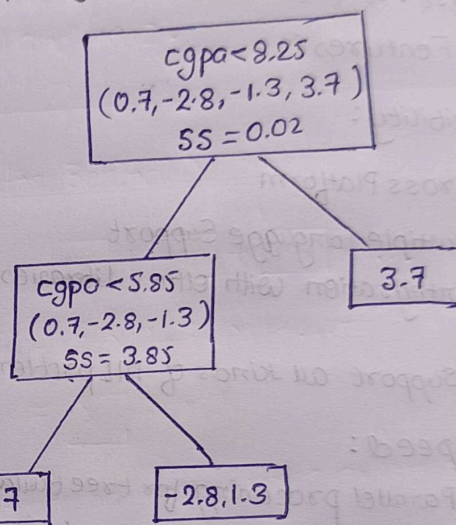
Clearly, the gain is maximum with splitting criteria of 8.25.

Thus, in stage 2 decision tree's root node will be $cgpa < 8.25$.



Now we will try to split this node
So, we have two splitting criteria now.

① splitting criteria $\rightarrow 5.85$

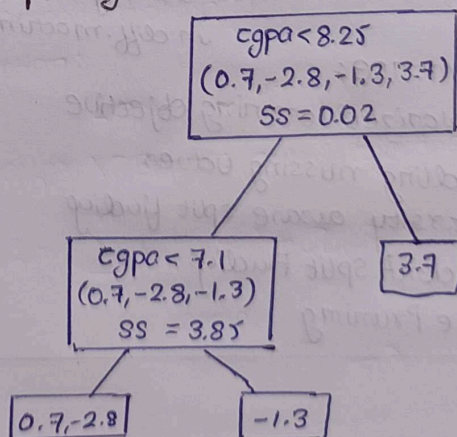


$$\therefore \text{gain} = SS_L + SS_R - SS_{\text{root}}$$

$$= \frac{(0.7)^2}{1} + \frac{(-2.8)^2 + (-1.3)^2}{2} - SS_R$$

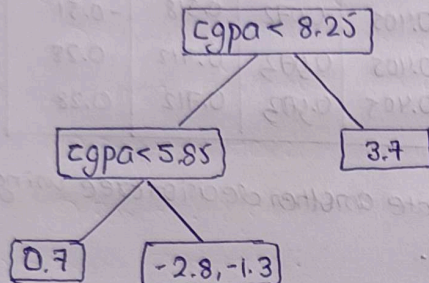
$$= 5.04$$

② splitting criteria $\rightarrow 7.1$



$$\begin{aligned}\% \text{ gain} &= SS_L + SS_R - SS_{\text{root}} \\ &= \frac{(0.7 - 2.8)^2}{2} + \frac{(-1.3)^2}{1} - 3.85 \\ &= 0.04\end{aligned}$$

Clearly, gain is more for splitting 5.85. We will not go further to split as it may lead to overfitting. So, our final decision tree is:



Now, we will calculate output of each leaf node. where,

$$\text{Output value} = \frac{\text{sum of residuals}}{\text{No. of residuals} + \lambda}$$

$$\text{where, } \lambda = 0$$

$$\therefore 0.7 \rightarrow \frac{0.7}{1} = 0.7$$

$$-2.8, -1.3 \rightarrow \frac{-2.8 - 1.3}{2} = -2.05$$

$$3.7 \rightarrow \frac{3.7}{1} = 3.7$$

So, the dataset be updated using the decision tree, and stage 1 model:

cgpa	package	model	residual	model2	res ₂
6.7	4.5	7.3	-2.8	6.69	-2.19
9.0	11.0	7.3	3.7	8.41	2.59
7.5	6.0	7.3	-1.3	6.69	-0.69
5.0	8.0	7.3	0.7	7.51	0.49

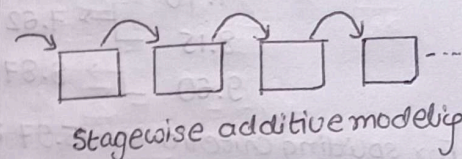
$$\uparrow$$

$$\text{mean} + \frac{1}{2} \times \frac{1}{0.3}$$

XGBoost for Classification

⇒ Let the dataset be:

cgpa	Placed?
5.70	0
6.25	1
7.10	0
8.15	1
9.60	1



Stage 1: we will find base estimator. We take log(odds) of output column as it's binary.

$$\log(\text{odds}) = \log\left(\frac{P}{1-P}\right) = \log\left(\frac{3/5}{2/5}\right) = 0.405$$

This 0.405 be same for each row as stage 1 prediction.

While in terms of probability,

$$\text{Prob.} = \frac{e^{\log \text{odds}}}{1 + e^{\log \text{odds}}} = \frac{e^{0.405}}{1 + e^{0.405}} = 0.60$$

So, the dataset now be:

cgpa	placed?	pred1	Pred1 (Prob)
5.70	0	0.405	0.6
6.25	1	0.405	0.6
7.10	0	0.405	0.6
8.15	1	0.405	0.6
9.60	1	0.405	0.6

Now, we will calc. error for this Stage1 model, called pseudo residual. Then we will create decision tree using the cgpa and resi column.

So, data now be:

cgpa	resi
5.70	-0.6
6.25	0.4
7.10	-0.6
8.15	0.4
9.60	0.4

We create one leaf node and find similarity score of them.

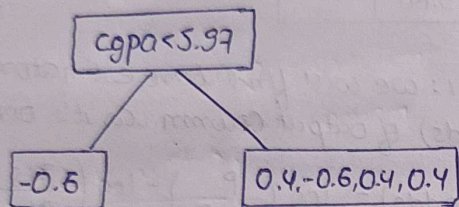
$$-0.6, 0.4, -0.6, 0.4, 0.4$$

$$\begin{aligned}SS &= \frac{(\sum \text{residual}_i)^2}{\sum (\text{Prev. prob}_i (1 - \text{Prev. prob}_i)) + \lambda} \\ &= \frac{(-0.6 + 0.4 - 0.6 + 0.4 + 0.4)^2}{5 \times [0.6(1 - 0.6)] + 0}\end{aligned}$$

Now, we will find the splitting criteria :

5.70	→ 5.97	} 4 splitting criteria
6.25	→ 6.67	
7.70	→ 7.62	
8.15	→ 8.87	
9.60	→ 8.87	

For splitting criteria → 5.97 :



$$SS_L = \frac{(-0.6)^2}{0.6(1-0.6)}$$

$$= 1.5$$

$$SS_R = \frac{(0.4-0.6+0.4+0.4)^2}{4 \times 0.6 \times 0.4}$$

$$= 0.37$$

$$\% \text{ gain} = (SS_L + SS_R) - SS_{\text{root}}$$

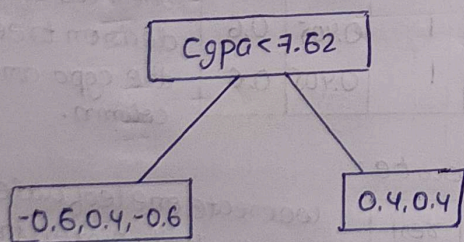
$$= (1.5 + 0.37) - 0$$

$$= 1.87$$

ily, gain for splitting criteria of 6.67 is 0.13.

ily " " " " of 7.62 is 2.22

% gain of 2.22 is maximum. So, the decision tree of stage 2 be:



$$\% \text{ output} = -1.11$$

$$\text{output} = 1.66$$

$$\% \text{ output} = \frac{\sum \text{residual}_i}{\sum (\text{pre-prob}_i (1 - \text{pre-prob}_i)) + \lambda}$$

Now, we will do prediction of combined model:
 $m_1 + 0.3 \times m_2$
 and then find its probability, and then residual2.
 So, the dataset be :

cgpa	placed?	pred1	pred2	Pre2 (prob)	res2
5.70	0	0.405	0.072	0.518	-0.51
6.25	1	0.405	0.072	0.518	0.48
7.10	0	0.405	0.072	0.518	-0.51
8.15	1	0.405	0.072	0.712	0.28
9.60	1	0.405	0.503	0.712	0.28

Next, we will create another decision tree using 'cgpa' and 'res2'.

Maths Behind XGBoost

$$\Rightarrow m_1 + m_2 + m_3 + \dots + m_n$$

$$y_i = f_1(x_i) + f_2(x_i) + \dots + f_n(x_i)$$

$$= \sum_{i=1}^n f_i(x_i)$$

% XGBoost basically works as an additive model.

⇒ output value of each leaf node :

$$\text{Output} = \frac{\sum \text{sum of residuals}}{\text{No. of residuals} + \lambda}$$

⇒ Loss function : relation b/w y and $\hat{y}_i$
 For Gradient Boosting it be any differentiable function :

$$L_{\text{on}} = L(y_i, \hat{y}_i)$$

while for XGBoost : (called objective function)

$$L = \sum_{i=1}^n L(y_i, \hat{y}_i) + \Omega(f_2(x_i))$$

where,

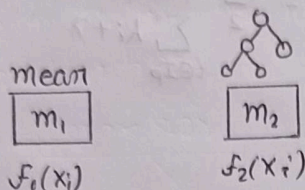
$$\Omega(f) = \gamma^T + \frac{1}{2} \lambda \|w\|^2$$

$T = \text{no. of leaf node}$; $\gamma, \lambda = \text{regularization parameters}$

⇒ Suppose we just use only one model in the XGBoost, for that the objective function will be

$$L^{(1)} = \sum_{i=1}^n L(y_i, \hat{y}_i)$$

Now, for next case, we take two models $f_1(x_i)$ and $f_2(x_i)$.



$$\hat{y}_i = f_1(x_i) + f_2(x_i)$$

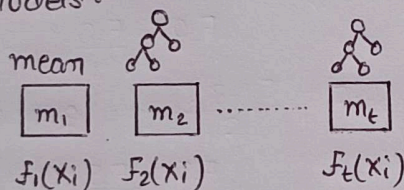
∴ Objective function here be:

$$L^{(2)} = \sum_{i=1}^n L(y_i, f_1(x_i) + f_2(x_i)) + \Omega(f_2(x_i))$$

Now, for next case we will have three models. For it the objective function be:

$$L^{(3)} = \sum_{i=1}^n L(y_i, f_1(x_i) + f_2(x_i) + f_3(x_i)) + \Omega(f_3(x_i))$$

Now, for t models:



∴ Objective functions for this:

$$L^{(t)} = \sum_{i=1}^n L(y_i, f_1(x_i) + f_2(x_i) + \dots + f_t(x_i)) + \Omega(f_t(x_i))$$

$$= \sum_{i=1}^n L(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t(x_i))$$

∴ at any stage t :

$$L^{(t)} = \sum_{i=1}^n L(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t(x_i))$$

Now, we want to minimize this objective function:

∴ because of decision tree, the objective function is not smooth and thus can't be differentiable.

∴ we will change the equation and approximate, we will do it by Taylor Series.

∴ Taylor series for $f(x)$ at any point a ,

$$f(x) \approx f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$

Now, for our objective function

$$L^{(t)} = \sum_{i=1}^n L(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t(x_i))$$

For this, the

$$x = \hat{y}_i^{(t-1)} + f_t(x_i)$$

$$a = \hat{y}_i^{(t-1)}$$

$$\therefore f(a) = \sum_{i=1}^n L(y_i, \hat{y}_i^{(t-1)})$$

$$\therefore f'(a)(x-a) = \sum_{i=1}^n \frac{\partial L(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)}} f_t(x_i)$$

$$\text{If } \frac{f''(a)(x-a)^2}{2} = \frac{1}{2} \sum_{i=1}^n \frac{\partial^2 L(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)^2}} f_t^2(x_i)$$

$$\therefore L^{(t)} = \sum_{i=1}^n L(y_i, \hat{y}_i^{(t-1)}) + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) + \Omega(f_t(x_i))$$

$$\Rightarrow L^{(t)} = \sum_{i=1}^n [g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)] + \Omega(f_t(x_i))$$

$$\Rightarrow L^{(t)} = \sum_{i=1}^n [g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)] + \gamma T + \frac{1}{2} \lambda [\omega_1^2 + \omega_2^2 + \dots + \omega_T^2]$$

$$\Rightarrow L^{(t)} = \sum_{i=1}^n [g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T \omega_j^2$$

$$\Rightarrow L^{(t)} = \sum_{j=1}^T \left[\sum_{i \in I_j} g_i \omega_j + \frac{1}{2} \sum_{i \in I_j} h_i \omega_j^2 \right] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T \omega_j^2$$

$$\Rightarrow L^{<t>} = \sum_{j=1}^T \left[\sum_{i \in I_j} g_i \omega_j + \frac{1}{2} \left(\sum_{i \in I_j} h_i + \lambda \right) \omega_j^2 \right] + \gamma T$$

$$\therefore \frac{\partial L}{\partial \omega_j} = \sum_{i \in I_j} g_i + \left(\sum_{i \in I_j} h_i + \lambda \right) \omega_j = 0$$

$$\Rightarrow \omega_j = \frac{- \sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda}$$

where, g_i = gradient
 h_i = hessian

$$\therefore g_i = \frac{\partial L(y_i, \hat{y}_i^{<t-1>})}{\partial \hat{y}_i^{<t-1>}}$$

$$h_i = \frac{\partial^2 L(y_i, \hat{y}_i^{<t-1>})}{\partial \hat{y}_i^{<t-1>2}}$$

For regression,

$$L = \frac{1}{2} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

$$\therefore g_i = \frac{\partial L}{\partial \hat{y}_i} = \sum_{i=1}^m (y_i - \hat{y}_i)$$

$$\therefore h_i = \frac{\partial^2 L}{\partial \hat{y}_i^2} = 1$$

$$\therefore \omega_j = \frac{\sum_{i \in I_j} (y_i - \hat{y}_i^{<t-1>})}{\sum_{i \in I_j} 1 + \lambda} = \frac{\text{sum of residuals}}{\text{No. of residuals} + \lambda}$$

For classification,

$$\omega_j = \frac{\text{sum of residuals}}{\sum_{i \in I_j} P_i(1 - P_i) + \lambda}$$

where, P_i = pred. prob. of
 previous iteration

Putting value of ω_j of regression in $L^{<t>}$:

$$L^{<t>}(q) = \frac{1}{2} \sum_{i=1}^T \frac{\left(\sum_{i \in I_j} g_i \right)^2}{\sum_{i \in I_j} h_i + \lambda} + \gamma T$$

For any leaf node,

$$L^{<t>}(q_p) = \frac{1}{2} \frac{\left(\sum_{i \in I_p} g_i \right)^2}{\sum_{i \in I_p} h_i + \lambda} + \gamma$$